University of Waterloo | Systems Design Engineering

User Study Evaluation Protocol: XAI Dashboards for Insider Threat Detection

Experimental Design & User Study Manual

Institution:	University of Waterloo
Department:	Systems Design Engineering
Prepared By:	Abdul Muqtadir Abbasi, PhD Candidate & Graduate Research Assistant
Supervised By:	Dr. Shi Cao
Co-Investigator:	Fariha Muqtadir, PhD Candidate
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Sample Size:	N=40 Participants (Equal stratification of University Proxy Students and Cybersecurity Industry Practitioners)

1. Research Framework

1.1 Research Question (RQ)

How does the integration of SHAP-based feature attribution visualizations (local waterfall and global beeswarm plots) into an anomaly detection dashboard affect a SOC analyst's efficiency, diagnostic accuracy, and cognitive workload when triaging multi-day insider threat alerts and discarding false alarms, compared to traditional raw log and Z-score analysis?

1.2 Core Evaluation Hypotheses

- **Hypothesis 1 (Efficiency & Speed):** Participants utilizing Design B (SHAP assisted) will correctly identify Confirm Threat instances and discard false alarms significantly faster than those utilizing Design A (Traditional baseline).
- **Hypothesis 2 (Diagnostic Accuracy):** Design B will maintain or improve classification accuracy by explicitly highlighting malicious indicators, minimizing human oversight under tight operational time constraints.
- **Hypothesis 3 (Cognitive Load Reduction):** Design B will significantly mitigate the mental demand and frustration metrics experienced by analysts by removing the requirement to manually correlate cross-sectional log indices and Z-score vectors.

2. Experimental Variables & Dashboard Architectures

This user study evaluates the performance impact of Explainable AI (XAI) overlays on standard anomaly monitoring workflows. It contrasts two distinct dashboard iterations:

- **Design A (Traditional Baseline Workflow):** Incorporates a standard temporal reconstruction error anomaly plot (with an explicit threat threshold line), a complete

tabular profile of input features paired with calculated Z-scores, and a deep-dive interactive panel displaying raw system event logs and high-level behavioral indicators.

- **Design B (Proposed XAI Workflow):** Preserves the core structural workflow of Design A (retaining the foundational anomaly plot, summary indicators, and logs to ensure baseline continuity) but embeds two critical SHAP overlays generated by a Random Forest surrogate model that explicitly leaves out the Removable Drive and composite Threat Activity flags:
  - **SHAP Beeswarm Plot:** Displays global historical feature contributions, illustrating average feature importance vectors across all audited days.
  - **SHAP Waterfall Plot:** Displays localized, single-day feature attributions for any selected flagged day, mapping precisely how positive feature forces pushed the Autoencoder's reconstruction error across the alert threshold line.

Methodological Note on Tool Isolation: In Design B, the traditional analysis components remain accessible to match ecological validity in operational deployment. The explicit objective is to demonstrate that the addition of SHAP plots changes the analyst's triage strategy—enabling rapid decision-making directly via the feature attribution plots without requiring exhaustive manual investigation of raw logs.

3. Participant Selection, Expertise Stratification & Remuneration

To evaluate if the proposed dashboard bridges the gap between expert and novice practitioners, $N = 40$ participants will be formally screened and categorized into two distinct expertise cohorts using a pre-screening intake metric:

- **Industry SOC Analyst Cohort ($n = 20$):** Comprises cybersecurity industry practitioners with active professional experience (e.g., SOC Analyst, SecOps, Penetration Testing).
- **University Student Cohort ($n = 20$):** Comprises university proxy students lacking professional defensive infrastructure experience.

3.1 Remuneration Protocol

To compensate for the 120-minute protocol, participants will be remunerated strictly according to their expertise cohort:

- **University Student Cohort:** CAD 15 per hour (Total: CAD 30 for the full 2-hour session, pro-rated at CAD 5 per 20 minutes of participation).
- **Industry SOC Analyst Cohort:** CAD 30 per hour (Total: CAD 60 for the full 2-hour session, pro-rated at CAD 10 per 20 minutes of participation).

4. Experimental Design & Balanced Allocation Matrix

The study employs a Within-Subject Counterbalanced Design. Every participant experiences both interfaces to control for individual variability, but they evaluate completely different users across the sessions to entirely eliminate the Learning Effect (Carryover Bias).

The study incorporates 4 foundational attack scenarios derived from the CERT r5.2 benchmarking dataset. To keep data distinct, each scenario features two separate user profiles exhibiting identical threat profiles but possessing completely unique timelines, values, and distinct flagged days:

Table 1: Study Configuration and Counterbalancing Matrix

PARTICIPANT ALLOCATION	**SCENARIO 1**	**SCENARIO 2**	**SCENARIO 3**	**SCENARIO 4**
Group 1 ($n = 20$) (10 Students + 10 Industry)	User 1A Design A (Baseline)	User 2B Design B (SHAP)	User 3A Design A (Baseline)	User 4B Design B (SHAP)
Group 2 ($n = 20$) (10 Students + 10 Industry)	User 1B Design B (SHAP)	User 2A Design A (Baseline)	User 3B Design B (SHAP)	User 4A Design A (Baseline)

5. Operational Testing Protocol

To ensure all participants are trained comprehensively and evaluated robustly, each active testing session is allocated a strict **120-minute block** and proceeds through three synchronized phases:

Phase 1: Targeted Standardization & Onboarding (60 Minutes)

- Execution of UW Ethics-approved institutional consent sheets (5 Minutes).
- Role assignment briefing: ”You are operating as a Proxy SOC Analyst. Your explicit objective is to audit 5 flagged operational days per user to isolate threat instances from malicious profiles and confidently dismiss false positive alarms.” (5 Minutes).
- Dashboard training: Detailed explanation of reconstruction error plots for both systems, followed by an extended instruction set explaining SHAP mechanics (Red blocks represent positive forces increasing anomaly score vectors; Blue blocks pull scores down) accompanied by guided practice (50 Minutes).

CRITICAL TRAINING CONSTRAINT (CASE CLOSURE DROPDOWN OPTIONS): Participants must be explicitly instructed during this onboarding phase that the ”Mixed Scenario (S5)” option visible within the case closure dropdown menu is an artifact of the legacy interface design and constitutes a wrong selection for this study. Analysts are strictly required to make a definitive operational choice (Confirm Threat vs. False Alarm). They must not select S5 as an uncertainty marker if they are confused; they must commit to a binary operational choice.

Phase 2: Execution & Structured Task Flow (50 Minutes Total)

Participants move through the 4 scenarios sequentially. This optimized budget allows exactly 12.5 minutes per scenario, reflecting roughly 2.5 minutes of dedicated processing time per anomaly vector.

- The internal logging engine triggers a silent timer immediately upon dashboard render.
- The analyst progresses through Day 1 to Day 5, declaring for each individual flag whether it constitutes a Confirm Threat (Malicious Threat) or a False Alarm (Benign Deviation).

- Immediately upon completing a scenario, the participant completes a localized, 1-page NASA Task Load Index (NASA-TLX) multi-axis assessment sheet to log subjective performance and cognitive strain.

Phase 3: Post-Study Quantitative Metrics (10 Minutes)
A final retrospective survey is executed to collect comprehensive interface feedback, incorporating a standardized System Usability Scale (SUS) questionnaire along with formal Likert confidence markers mapping tool trust.

6. Quantitative Metrics & Statistical Analysis Blueprint

The system records extensive trial metrics to generate data points suitable for academic publication and thesis defense criteria (40 analysts x 4 scenarios x 5 days = 800 granular triage vectors):

- **Time-to-Decision (TtD):** Evaluated using a Two-Way Mixed ANOVA to isolate main interaction effects across the primary interface type (Design A vs. B) and user expertise level (Student vs. Industry).
- **Triage Accuracy:** Assessed via binary correctness logs mapping the accuracy delta between the two interface profiles.
- **Subjective Workload (NASA-TLX):** Evaluated via paired non-parametric tests (Wilcoxon Signed-Rank) contrasting workload variance scores.